

From Clusters to Views: Entity-Centric Functional Aggregation of Crowdsourced Test Reports

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CCS Concepts

• Software and its engineering → Software testing and debugging.

Keywords

crowdsourced testing, bug report analysis, report aggregation, software testing

1 Detailed ECFA Entity Schema

The main paper introduces a compact ECFA schema and reserves the detailed operational definitions for the appendix. Table 1 expands the six entity types used in ECFA and keeps the same extraction boundary as the main text. In particular, MOD denotes the functional locus, ELEM denotes a concrete interaction object within that locus, PHEN records user-observable failure manifestations, DIAG is limited to explicit system-presented messages, and ISSUE is a compact problem label rather than a free-form summary.

2 Stage 1 and Stage 2 Prompt Skeletons

This section reports the prompt skeletons used in the first two stages of ECFA. Table 2 summarizes the Stage 1 extraction prompt, and Table 3 summarizes the Stage 2 prompt family used in FSA. We keep the method-relevant constraints and output contracts and omit batching and logging details that do not affect the algorithmic description.

3 RQ1 Supplementary Results

Table 4 reports the per-app diagnostics used in RQ1. Besides the ratio-based quantities shown in the main paper, we also report $|V|$ and $|V_{LCC}|$ to make the denominator effect explicit, together with provenance-set statistics that reflect evidence-set inflation after unification.

4 Additional RQ2 Details

4.1 Supplementary comparison with generic external baselines

We additionally compare ECFA with three generic external baselines under the same induced candidate pool and the same shared review-cost budget: **MMR**, **FacilityLocation**, and **Louvain-Medoid**. Among them, **Louvain-Medoid** is the strongest competitor, but it still trails ECFA on the main coverage-under-budget metrics. **MMR** and **FacilityLocation** are clearly weaker under the same protocol. The full comparison is reported in Table 5.

5 Additional RQ3 Details

5.1 LLM-assisted student annotation protocol for cross-cluster relevance

RQ3 uses adjudicated labels obtained under fixed clustering boundaries rather than a purely automatic silver signal.

Let $\mathcal{D}_A$ be the report corpus for an app A , and let $\text{cluster}_B(r)$ denote the cluster assignment of report r under a fixed boundary $B \in \{\text{TF-IDF+HC}, \text{SBERT+HC}, \text{LLMCluster}\}$. For a query report $q \in \mathcal{D}_A$, the cross-cluster search space is

$$\mathcal{U}_B(q) = \{r \in \mathcal{D}_A \setminus \{q\} \mid \text{cluster}_B(r) \neq \text{cluster}_B(q)\}. \quad (1)$$

Because exhaustive adjudication over $\mathcal{U}_B(q)$ is impractical, we build for each query-boundary unit a manageable annotation pool directly from the raw reports under the fixed boundary. This step is independent of ECFA: annotators do not see extraction outputs, unified entities, induced views, or retrieval rankings from ECFA. For each candidate pair (q, r) with $r \in \mathcal{U}_B(q)$, the LLM provides only a preliminary relevance suggestion and a short rationale.

Three student annotators then review each candidate pair together with the LLM suggestion and decide whether the pair is relevant to the query under boundary B . Final labels are determined by majority vote. The adjudicated cross-cluster relevant set for query q under boundary B is

$$\text{Rel}_B^{\text{cross}}(q) = \left\{ r \in \mathcal{U}_B(q) \mid \begin{array}{l} (q, r) \text{ is accepted as relevant} \\ \text{by majority vote} \end{array} \right\}. \quad (2)$$

The main RQ3 benchmark is therefore a comparative retrieval evaluation over an adjudicated candidate pool under fixed boundaries. All main RQ3 metrics are computed against these adjudicated labels, and only adjudicated candidate pairs contribute labeled positives.

Secondary blinded audit subset. We additionally construct a smaller blinded audit subset for robustness checking. We first sample 120 query reports in a stratified manner across the 6 apps (20 per app), restricting sampling to queries that have cross-cluster relevant evidence under all three boundaries. Instantiating these queries under TF-IDF+HC, SBERT+HC, and LLMCluster yields 360 query-boundary units.

For each unit, we form a blinded judgment pool by taking the union of the top-20 cross-cluster candidates returned by ECFA-Retr, Global, and MultiCluster-L5, and then removing duplicates. Three student annotators review each pooled candidate blind to method identity and rank position, and majority vote again determines the final label. Because this subset is much smaller but more tightly controlled than the main benchmark, we use it as an audit set rather than as a replacement benchmark. On this subset, the paper reports Success@20, Recall@20, MRR@20, and nDCG@20. Recall-based metrics are computed against the adjudicated pooled candidate set for each sampled query-boundary unit rather than against the full corpus.

Table 1: (Appendix) Detailed ECFA entity schema. The four core types are strictly span-grounded; the two extended types may be constructed at the analysis level when missing, but must preserve provenance.

Type	Abbr.	Operational definition (for extraction/auditing)	Evidence policy
FunctionalModule	MOD	A named functional unit, page, or flow node in the app, typically anchored by UI text, navigation items, or functional entry descriptions (for example, "login page", "order submission", "payment", or "activity record").	Span-grounded only: must be traceable to a source span; synonym normalization may consolidate mentions but must retain provenance pointers.
UserAction	OP	An observable user action together with its target, capturing the triggering interaction under the local functional context (for example, "tap ...button", "swipe the list", "enter verification code", or "go back"). Action verbs are normalized to a small action inventory, while the target keeps the source-facing surface form.	Span-grounded only: must be traceable to source text; no hallucinated actions or targets are allowed.
ObservedSymptom	PHEN	A user-observable abnormal outcome that deviates from expected behavior (for example, "page frozen", "crash", "button unresponsive", or "empty/incorrect display"). Original phrasing is preserved as much as possible to avoid premature reinterpretation.	Span-grounded only: must be traceable; only denoising or synonym normalization is allowed, with no causal inference.
SystemDiagnosis	DIAG	An explicitly presented diagnostic signal, such as an error code, toast/dialog message, or the first line of an exception (for example, "Error 500", "network connection failed", or "NullPointerException"). Only user-visible information is recorded, without backend speculation.	Span-grounded only: must be traceable; if no explicit diagnostic signal is present, DIAG is absent rather than inferred.
ImpactElement	ELEM	A concrete affected UI element or interaction object (for example, "login button", "order list item", "avatar upload control", or "step-count card"). This type refines where a symptom or diagnosis lands inside a functional locus and supports finer-grained view organization.	Prefer span-grounded: when no explicit mention exists, allow deterministic analysis-level construction from grounded OP/PHEN/DIAG context, while preserving source nodes, edges, and evidence pointers.
ProblemStatement	ISSUE	A compact defect statement used to unify coverage accounting and provide a stable view name (for example, "login failure", "payment flow crash", or "activity data not synced"). It may be induced from recurring PHEN/DIAG evidence together with contextual MOD/OP information.	Prefer span-grounded: when missing, allow deterministic provenance-preserving construction from grounded PHEN/DIAG plus contextual MOD/OP, but not as a free-form abstract summary.

Note. This detailed schema expands the compact table in the main paper without changing the type inventory or evidence-policy boundary. It is also consistent with the paper's extraction policy: unsupported completion is disallowed, and retained relation types remain limited to contextual anchoring, interaction-target, outcome, and support/derivation links.

Table 2: Stage 1 extraction prompt skeleton used in ECFA.

Block	Prompt skeleton
Goal	Extract only schema-legal evidence triples that are explicitly supported by the report text. Do not infer missing causes, complete unstated details, or rewrite the report into an abstract summary. Every retained item must remain auditable against the source text.
Allowed entity types	Use only the six ECFA entity types: FunctionalModule (MOD), UserAction (OP), ImpactElement (ELEM), ObservedSymptom (PHEN), SystemDiagnosis (DIAG), and ProblemStatement (ISSUE).
Entity rules	MOD: only named pages, modules, flows, settings, rules, or other functional loci explicitly mentioned in the report. OP: only explicit user-initiated actions, including explicit "cannot/failed to X" patterns where X is itself a user action. ELEM: only concrete controls or interaction objects explicitly mentioned in the report; light normalization is allowed only when the object is clearly present in the text. PHEN: only user-observable abnormal outcomes, using wording close to the report. DIAG: only explicit user-visible diagnostic signals, such as error codes, dialog text, toast text, or exception strings. ISSUE: only compact, evidence-grounded problem statements; no free-form summaries or causal explanations.
Allowed relations	Only the following relations are allowed: located_in (ELEM→MOD), performs_on (OP→ELEM), triggers (OP→MOD), results_in (OP→PHEN), accompanied_by (PHEN↔DIAG), reported_in (DIAG→MOD), and concerns (ISSUE→MOD/ELEM).
Exclusion rule	Unsupported completion is disallowed. If an entity or relation is not directly supported strongly enough by the report text, omit it.
Output format	Return valid JSON only. The top level is an array, one object per report: {"report_id": "...", "triples": [...]}. Each triple contains subject, subject_type, relation, object, and object_type. If a report has no valid triple, return an empty triples array.
Post-checks	After generation, EAL applies type-legality checks, relation-compatibility checks, span recoverability checks, and deduplication. The prompt is therefore only the first acceptance layer; final admissibility is determined jointly by prompt constraints and deterministic filtering.

Note. The table reports the stable extraction skeleton used in Stage 1.

Table 3: Stage 2 prompt family used for type-constrained entity unification.

FSA step	Prompt skeleton
Step 1: coarse semantic bucketing	Group same-type ECFA entities into coarse concept buckets and provide a short operational definition for each bucket. Do not merge incompatible semantics or different functional roles. Keep page/module loci distinct from concrete UI elements, user actions distinct from symptoms or problem statements, and coarse flows distinct from local controls. Return a JSON mapping from bucket ID to concept, definition, scope_keywords, and examples.
Step 2: pairwise equivalence scoring within a bucket	Judge whether two same-type entities inside the same coarse bucket are semantically equivalent. Type must match, target object must match when applicable, and local meaning must overlap. Surface-form variation is allowed, but page-vs-dialog differences, control-vs-process differences, and clear granularity mismatches lower the score or block equivalence. Return a JSON array of [entityA, entityB, score, reason, type]. The scoring rubric is strict: 0.9 = essentially equivalent; 0.7 = plausible variant but not fully stable; 0.5 = related but not equivalent.
Step 3: batch veto check	Re-check candidate equivalent pairs in batch and veto pairs that violate type consistency, object consistency, medium consistency, granularity consistency, or semantic-scope overlap. Return a JSON object mapping each pair ID to either "pass" or "veto".
Acceptance rule	Alignment is allowed only within the same entity type. Decisions use anchored mentions, provenance snippets, and local same-type evidence rather than surface-form similarity alone. Uncertain pairs remain unmerged.
Deterministic closure	After prompt-based validation, the pipeline applies the fixed thresholds reported in the main paper, keeps accepted same-type links, and performs connected-component closure to obtain canonical representatives. Canonical names are assigned from the most frequent normalized surface form in each accepted component, while source provenance is preserved.

Note. Thresholding, candidate generation, and closure are deterministic pipeline steps outside the language model.

5.2 Raw records for the preliminary human-task audit

The main paper reports condition-level aggregates for the pilot human-task audit under LLMcluster. Table 6 gives the underlying

Table 4: (Appendix) RQ1 per-App diagnostics: denominator effect, continuous coupling, and evidence-set inflation. Each cell shows $w/o \rightarrow w/(\Delta)$.

App	$ V $ (abs)	$ V_{LCC} $ (abs)	$\bar{d}_J$ ↑	$Avg \Gamma(v) $ ↑	$Avg \Gamma(u) \cup \Gamma(v) $ ↑	$Avg \Gamma(u) \cap \Gamma(v) $ ↑
Slife	4052→3913 (-139)	2165→2248 (+83)	0.507→0.532 (+0.025)	1.472→1.523 (+0.051)	9.719→11.921 (+2.202)	1.121→1.138 (+0.016)
JayMe	2237→1924 (-313)	1321→1378 (+57)	0.525→0.610 (+0.085)	1.493→1.733 (+0.241)	8.827→14.143 (+5.316)	1.132→1.198 (+0.066)
Huawei Health	872→790 (-82)	271→393 (+122)	0.299→0.401 (+0.102)	1.212→1.334 (+0.122)	2.976→4.188 (+1.212)	1.062→1.098 (+0.036)
Hujiang English	1054→944 (-110)	493→580 (+87)	0.357→0.445 (+0.088)	1.308→1.457 (+0.148)	3.394→4.426 (+1.032)	1.088→1.127 (+0.039)
Tuniu	7680→7340 (-340)	4345→4447 (+102)	0.531→0.561 (+0.030)	1.465→1.531 (+0.066)	20.280→20.802 (+0.522)	1.086→1.103 (+0.017)
IELTS Listening	898→805 (-93)	110→335 (+225)	0.294→0.402 (+0.108)	1.249→1.386 (+0.137)	2.495→3.187 (+0.692)	1.090→1.121 (+0.031)

Note. $Avg|\Gamma(u) \cup \Gamma(v)|$ and $Avg|\Gamma(u) \cap \Gamma(v)|$ are computed over eligible edges E' (endpoints non-empty).

Table 5: Supplementary comparison with generic external baselines for RQ2 under the same induced candidate pool and the same shared review-cost budget. Higher is better for nAUC@B, Cov@0.50B, Cov@0.75B, and Sel@B; lower is better for C/V.

Method	nAUC@B	Cov@0.50B	Cov@0.75B	Sel@B	C/V
ECFA	0.451	0.513	0.656	50.0	25.272
Louvain-Medoid	0.359	0.402	0.549	47.8	14.458
MMR	0.318	0.369	0.522	12.2	92.205
FacilityLocation	0.225	0.252	0.308	41.7	2.864

Note. Values are averaged across the 6 apps. Louvain-Medoid is the strongest of the three external baselines, but it remains below ECFA on the main budgeted coverage metrics.

12 student-app units for the four audited apps (Huawei Health, Hujiang English, JayMe, and IELTS Listening). The mean row recovers the condition-level values reported in the main paper after rounding: LLMcluster (31.50 min, 0.613 Theme-F1, 0.442 Support-F1, 0.50 Hit@2) and ECFA (25.58 min, 0.707 Theme-F1, 0.555 Support-F1, 0.75 Hit@2), with ECFA winning 10 of 12 units.

Table 6: Raw records for the preliminary human-task audit under LLMCluster (4 apps × 3 students = 12 student-app units). The mean row reproduces the condition-level aggregates reported in the main paper after rounding.

App	Stu.	LLMCluster				ECFA				Winner
		Time	Theme-F1	Support-F1	Hit@2	Time	Theme-F1	Support-F1	Hit@2	
Huawei Health	A	28	0.69	0.55	1	23	0.79	0.68	1	ECFA
Huawei Health	B	31	0.65	0.50	1	24	0.75	0.61	1	ECFA
Huawei Health	C	27	0.63	0.47	1	21	0.72	0.59	1	ECFA
Hujiang English	A	34	0.64	0.46	0	26	0.77	0.60	1	ECFA
Hujiang English	B	32	0.61	0.44	1	25	0.73	0.56	1	ECFA
Hujiang English	C	29	0.58	0.40	0	23	0.70	0.53	1	ECFA
JayMe	A	37	0.66	0.48	1	30	0.74	0.58	1	ECFA
JayMe	B	27	0.62	0.45	0	29	0.60	0.47	0	LLMCluster
JayMe	C	26	0.59	0.41	1	28	0.57	0.42	1	LLMCluster
IELTS Listening	A	39	0.60	0.42	0	28	0.76	0.61	1	ECFA
IELTS Listening	B	36	0.56	0.38	0	26	0.71	0.55	0	ECFA
IELTS Listening	C	32	0.52	0.34	0	24	0.64	0.46	0	ECFA
Mean		31.50	0.613	0.442	0.50	25.58	0.707	0.555	0.75	ECFA 10/12

Note. Theme-F1 and Support-F1 are computed against the reference themes and supporting evidence, respectively. Hit@2 is the fraction of units whose first two inspected items include a correct target. The winner column is assigned per student-app unit.